

TECHNOLOGY STACK

EduGenie: Google Gemini-Powered Learning Assistant

Date	07/07/2026	Maximum Marks	2 Marks
Team ID	XXXXXX	Project Name	EduGenie: Google Gemini-Powered Learning Assistant

Define Your Technology Stack: Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution.

A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements. Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component/ Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/ CSS)</i>	React.js & Tailwind CSS	React provides exceptional component reusability and a virtual DOM for a fast, responsive user experience. Tailwind CSS allows rapid, responsive UI development tailored for a seamless learning interface across device types.
2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Python (FastAPI)	FastAPI offers high performance, asynchronous request handling, and native Python support. This guarantees seamless integration with Google's AI libraries and fast JSON processing for real-time tutoring responses.

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3	Database / Data Storage (e.g., MySQL, MongoDB, PostgreSQL)	PostgreSQL & ChromaDB	PostgreSQL is utilized for robust relational mapping of user metrics, progression tracking, and logs. ChromaDB acts as a high-performance vector database to store and retrieve educational content embeddings for efficient RAG (Retrieval-Augmented Generation).
4	Cloud / Hosting / Deployment (e.g., AWS, Azure, Heroku, Docker)	Google Cloud Platform (GCP) & Docker	Containerization with Docker ensures environment consistency. Deploying via GCP (Cloud Run) offers native integration with Google ecosystem tools, cost-effective serverless hosting, and immediate horizontal scalability.
5	Version Control & CI/CD (e.g., GitHub, GitLab, Jenkins)	GitHub & GitHub Actions	Enables structured branch management and seamless team collaboration. GitHub Actions provides automated testing and continuous integration pipelines to build and deploy Docker containers directly into Google Cloud.
6	Third-Party APIs / Other Tools (e.g., Stripe, Twilio, Google Maps)	Google Gemini API & Firebase Auth	The Gemini API serves as the primary cognitive engine for generating customized quizzes, summaries, and answering learner queries. Firebase Auth delivers reliable, secure, and ready-to-use user authentication.